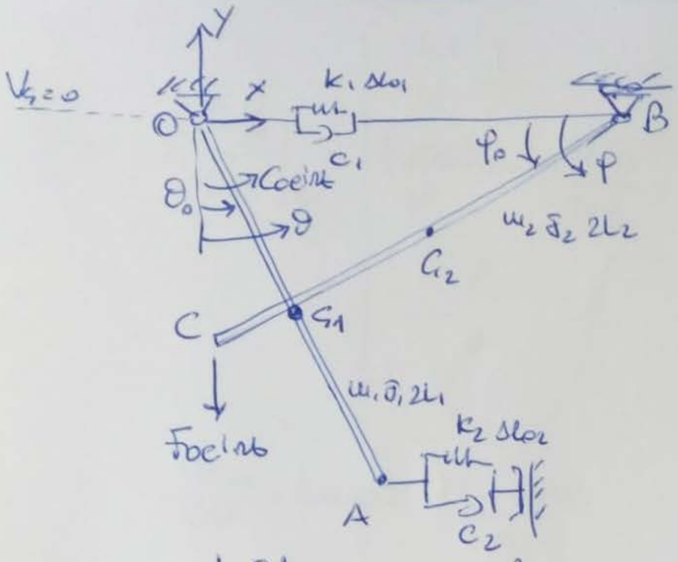
WRITTEN TEST - SEPTEMBER 01ST 2021




- 2 rigid bodies : 6 dof
- 1 hinge : 2 doc.
- 1 cart : 1 d.o.c.
- constraint in $G_1$ equivalent to a cart because it allows for a relative rotation and translation of $\overline{BC}$ with respect to $\overline{OA}$ → 1 doc.

2 dof.

$$\underline{x} = \begin{Bmatrix} \vartheta \\ \varphi \end{Bmatrix} \quad \underline{x}_0 = \begin{Bmatrix} \vartheta_0 \\ \varphi_0 \end{Bmatrix}$$

$$T = \frac{1}{2} m_1 v_{G1}^2 + \frac{1}{2} J_1 \dot\vartheta^2 + \frac{1}{2} m_2 \dot x_{G2}^2 + \frac{1}{2} m_2 \dot y_{G2}^2 + \frac{1}{2} J_2 \dot\varphi^2$$

$$D = \frac{1}{2} \sum_{i=1}^{2} c_i \Delta \dot l_i^2 \;;\; V_{el} = \frac{1}{2} \sum_{i=1}^{2} k_i \Delta l_i^2 \;;\; V_g = \sum_{i=1}^{2} m_i g\, h_{Gi}$$

$$\delta W = C_0 e^{i\Omega t} \delta\vartheta + \vec{F_0} e^{i\Omega t} \times \delta \vec{S}_c = C_0 e^{i\Omega t} \delta\vartheta - F_0 e^{i\Omega t} \delta y_c$$

$$[m_{PH}] = \mathrm{diag}(m_1\ J_1\ m_2\ m_2\ J_2) \;;\; \underline{y}_m = \{ S_{G1}\ \vartheta\ x_{G2}\ y_{G2}\ \varphi \}^T$$

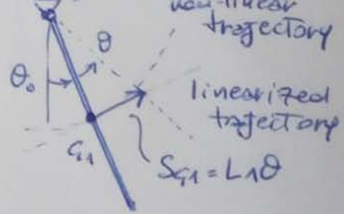
$$[C_{PH}] = \mathrm{diag}(c_1\ c_2) \;;\; [k_{PH}] = \mathrm{diag}(k_1\ k_2) \;;\; \underline{y}_c = \underline{y}_k = \{ \Delta l_1\ \Delta l_2 \}^T$$

$$\underline{F} = \{ C_0 e^{i\Omega t}\ F_0 e^{i\Omega t} \}^T \;;\; \underline{y}_Q = \{ \vartheta\ \ -y_c \}^T$$

- Non Linear kinematic relationships

$$x_{G1} = L_1 \sin\vartheta \;;$$

$$\left.\begin{aligned} y_{G1} &= -L_1 \cos\vartheta \\ \overline{BG_1} &= \xi \to y_{G1} = -\xi \sin\varphi \end{aligned}\right\} \to \xi = L_1 \frac{\cos\vartheta}{\sin\varphi}$$




$$x_B = L_1 \sin\vartheta + \xi \cos\varphi = L_1 \sin\vartheta + L_1 \frac{\cos\vartheta}{\tan\varphi}$$

$$x_{G2} = L_1 \sin\vartheta + \frac{L_1 \cos\vartheta}{\tan\varphi} - L_2 \cos\varphi$$

$$y_{G2} = -L_2 \sin\varphi$$

$$x_A = 2L_1 \sin\varphi \;;\; y_c = -2L_2 \sin\varphi$$

$$\Delta l_1 = x_B - x_{B0}$$

$$\Delta l_2 = -\Delta x_A = -(x_A - x_{A0})$$

$$[\Lambda_m] = \left.\frac{\partial \underline{Y}_m}{\partial \underline{X}}\right|_{\underline{X}_0} = \begin{bmatrix} L_1 & \phi \\ 1 & \phi \\ L_1\cos\vartheta_0 - L_1 \dfrac{\operatorname{sen}\vartheta_0}{\operatorname{tg}\varphi_0} & -\dfrac{L_1\cos\vartheta_0}{\operatorname{sen}^2\varphi_0} + L_2 \operatorname{sen}\varphi_0 \\ \phi & -L_2 \cos\varphi_0 \\ \phi & 1 \end{bmatrix}$$

$$\frac{\partial x_{G2}}{\partial \vartheta} = L_1\cos\vartheta - L_1 \frac{\operatorname{sen}\vartheta}{\operatorname{tg}\varphi} \qquad \frac{\partial x_{G2}}{\partial \varphi} = -\frac{L_1\cos\vartheta}{\operatorname{sen}^2\varphi} + L_2 \operatorname{sen}\varphi$$

NB: it's the same to define T as : $T = \underline{\underline{\frac{1}{2} m_1 \dot{x}_{G1}^2 + \frac{1}{2} m_1 \dot{y}_{G2}^2}} + \frac{1}{2} J_1 \dot{\vartheta}^2 + \frac{1}{2} m_2 \dots$

$\rightarrow [m_{PH}] = \operatorname{diag}(\underline{\underline{m_1 \;\; m_1}} \;\; J_1 \;\; m_2 \; \dots) \rightarrow$

$$\rightarrow \Lambda_m = \begin{bmatrix} L_1 \cos\vartheta_0 & 0 \\ L_1 \operatorname{sen}\vartheta_0 & 0 \\ 1 & 0 \\ L_1\cos\vartheta_0 - L_1 \dfrac{\operatorname{sen}\vartheta_0}{\operatorname{tg}\varphi_0} & - - \\ - - - & \end{bmatrix}$$

$$[\Lambda_k] = \begin{bmatrix} L_1\cos\vartheta_0 - L_1 \dfrac{\operatorname{sen}\vartheta_0}{\operatorname{tg}\varphi_0} & -L_1 \dfrac{\cos\vartheta_0}{\operatorname{sen}^2\varphi_0} \\ -2L_1 \cos\vartheta_0 & 0 \end{bmatrix} = [\Lambda_c]$$

$$[\Lambda_0] = \begin{bmatrix} 1 & 0 \\ 0 & 2L_2 \cos\varphi_0 \end{bmatrix}$$

$$
\begin{cases}
\dfrac{\partial^2 \Delta l_1}{\partial \theta^2} = -L_1 \operatorname{sen}\theta - L_1 \dfrac{\cos\theta}{Tg\varphi} \\[2ex]
\dfrac{\partial^2 \Delta l_1}{\partial \theta \partial \varphi} = L_1 \operatorname{sen}\theta \dfrac{1}{\operatorname{sen}^2\varphi} \\[2ex]
\dfrac{\partial^2 \Delta l_1}{\partial \varphi^2} = L_1 \cos\theta \ \dfrac{2\operatorname{sen}\varphi\cos\varphi}{\operatorname{sen}^4\varphi}
\end{cases}
\qquad
\left[k_{el\,II\,1}\right] = k_1 \Delta l_{01}
\begin{bmatrix}
-L_1 \operatorname{sen}\theta_0 - L_1 \dfrac{\cos\theta_0}{Tg\varphi_0} & \dfrac{L_1 \operatorname{sen}\theta_0}{\operatorname{sen}^2\varphi_0} \\[2ex]
\dfrac{L_1 \operatorname{sen}\theta_0}{\operatorname{sen}^2\varphi_0} & \dfrac{2L_1 \cos\theta_0 \cos\varphi_0}{\operatorname{sen}^3\varphi_0}
\end{bmatrix}
$$

$$
\frac{\partial^2 \Delta l_2}{\partial \theta^2} = 2L_1 \operatorname{sen}\varphi \rightarrow \left[k_{el\,II\,2}\right] = k_2 \Delta l_{02}
\begin{bmatrix}
2L_1 \operatorname{sen}\theta_0 & 0 \\
0 & 0
\end{bmatrix}
$$

$$
\begin{cases}
h_{G1} = y_{G1} \\
h_{G2} = y_{G2}
\end{cases}
\rightarrow \left[k_{G1}\right] = m_1 g
\begin{bmatrix}
L_1 \cos\theta_0 & 0 \\
0 & 0
\end{bmatrix}
; \ \left[k_{G2}\right] = m_2 g
\begin{bmatrix}
0 & 0 \\
0 & L_2 \operatorname{sen}\varphi_0
\end{bmatrix}
$$

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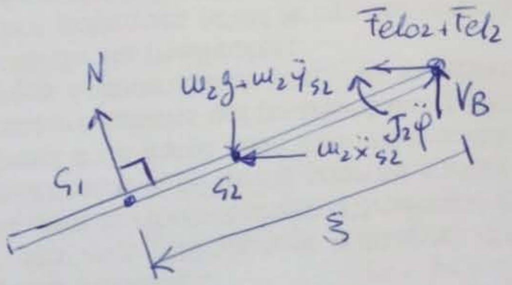
- questions 3 and 4: output $y_{G2}$, $x_B$, $X_{G2}$, $Fel_2$ <u>LINEARIZED</u>

$$Fel_2 = (k_2 + i\Omega c_2)\,\Delta l_{2,lin}$$

- question 5:




$$\Sigma M_B^{CB} = 0 \ : \quad N\xi - m_2 g L_2 \cos\varphi - m_2 \ddot{y}_{G2} L_2 \cos\varphi + m_2 \ddot{x}_{G2} L_2 \operatorname{sen}\varphi + J_2 \ddot{\varphi} = 0 \qquad (1)$$

$$N = N_{ST} + N_{dyn}$$

$$\xi = L_1 \frac{\cos\theta}{\operatorname{sen}\varphi} = f(\theta, \varphi)$$

$$\xi \simeq f(\theta_0, \varphi_0) + \left.\frac{\partial f}{\partial \theta}\right|_{\underline{x}_0} (\theta - \theta_0) + \left.\frac{\partial f}{\partial \varphi}\right|_{x_0} (\varphi - \varphi_0) = L_1 \frac{\cos\theta_0}{\operatorname{sen}\varphi_0} - L_1 \frac{\operatorname{sen}\theta_0}{\operatorname{sen}\varphi_0}(\theta - \theta_0) + $$
$$- L_1 \cos\theta_0 \frac{\cos\varphi_0}{\operatorname{sen}^2\varphi_0}(\varphi - \varphi_0)$$

$$m_2 g L_2 \cos\varphi \simeq m_2 g L_2 \cos\varphi_0 - m_2 g L_2 \operatorname{sen}\varphi_0 (\varphi-\varphi_0)$$

$$m_2 \ddot{y}_{s2} L_2 \cos\varphi \simeq m_2 \ddot{y}_{s2,lin} L_2 \cos\varphi_0 - \cancel{m_2 \ddot{y}_{s2,lin} L_2 \operatorname{sen}\varphi_0 (\varphi-\varphi_0)}$$

2nd order Term

$$m_2 \ddot{x}_{s2} L_2 \operatorname{sen}\varphi \simeq m_2 \ddot{x}_{s2,lin} L_2 \operatorname{sen}\varphi_0 + \cancel{m_2 \ddot{x}_{s2,lin} L_2 \cos\varphi_0 (\varphi-\varphi_0)}$$

2nd order Term

$$N\xi = \left(N_{ST} + N_{dyn}\right)\left(\xi_0 + \left.\frac{\partial \xi}{\partial \theta}\right|_{\underline{x}_0}(\theta-\theta_0) + \left.\frac{\partial \xi}{\partial \varphi}\right|_{\underline{x}_0}(\varphi-\varphi_0)\right) =$$

$$= \underbrace{N_{ST}\,\xi_0}_{\text{state Term}} + \underbrace{N_{ST}\left[\left.\frac{\partial \xi}{\partial \theta}\right|_{\underline{x}_0}(\theta-\theta_0) + \left.\frac{\partial \xi}{\partial \varphi}\right|_{\underline{x}_0}(\varphi-\varphi_0)\right] + N_{dyn}\,\xi_0}_{1^{st}\text{ order Term}} + \underbrace{N_{dyn}\left[\cancel{\left.\frac{\partial \xi}{\partial \theta}\right|_{\underline{x}_0}} \dots\right]}_{2^{nd}\text{ order Term}}$$

$$N_{ST}\,\xi_0 = m_2 g L_2 \cos\varphi_0 \rightarrow \boxed{N_{ST} = m_2 g \frac{L_2}{\xi_0}\cos\varphi_0}$$

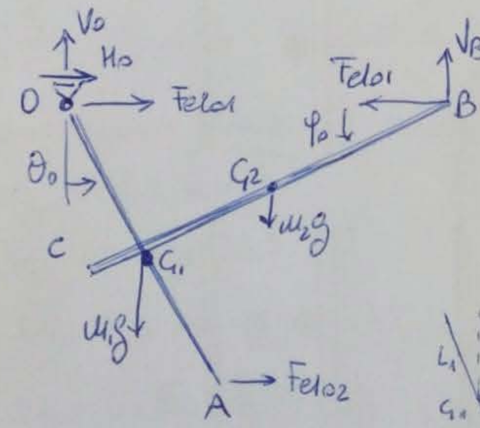
(from (1), equilibrium of the state Terms)

$$\boxed{\begin{aligned} N_{dyn}\,\xi_0 = {} & N_{ST} L_1 \frac{\operatorname{sen}\theta_0}{\operatorname{sen}\varphi_0}(\theta-\theta_0) + N_{ST} L_1 \frac{\cos\theta_0 \cos\varphi_0}{\operatorname{sen}^2\varphi_0}(\varphi-\varphi_0) + \\ & - m_2 g L_2 \operatorname{sen}\varphi_0 (\varphi-\varphi_0) + m_2 \ddot{y}_{s2,lin} L_2 \cos\varphi_0 - m_2 \ddot{x}_{s2,lin} L_2 \operatorname{sen}\varphi_0 - \bar{J}_2 \ddot{\varphi} \end{aligned}}$$

↓

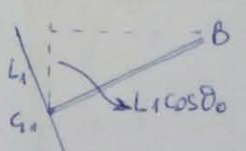
not considering the Terms depending on $N_{ST}$ and $m_2 g$, the solution was not considered wrong.

- Question 6




$$\xi_0 = \overline{BG_1} = L_1 \frac{\cos\theta_0}{\operatorname{sen}\varphi_0}$$

$$\Sigma M_{G_1}^{CB} = 0 \ : \ F_{elo1} L_1 \cos\theta_0 - m_1 g\,(\xi_0 - L_2)\cos\varphi_0 \rightarrow$$




$$\rightarrow \Delta l_{o1} = \frac{m_1 g}{K_1}\,\frac{\xi_0 - L_2}{L_1}\,\frac{\cos\varphi_0}{\cos\theta_0}$$

$$\Sigma M_0 = 0 \ : \ m_1 g L_1 \operatorname{sen}\theta_0 + m_2 g\left[(\xi_0 - L_2)\cos\varphi_0 + L_1 \operatorname{sen}\theta_0\right] - F_{elo2}\, 2L_1 \cos\theta_0 = 0 \rightarrow$$

$$\rightarrow \Delta l_{o2} = \frac{m_1 g L_1 \operatorname{sen}\theta_0 + m_2 g\left[(\xi_0 - L_2)\cos\varphi_0 + L_1\operatorname{sen}\theta_0\right]}{2L_1 k_2 \cos\theta_0}$$